

Token-Level Uncertainty as a Predictor of Hallucination in Large Language Models

Abstract

Hallucinations in large language models (LLMs) represent a critical challenge to their deployment in high-stakes applications. This paper investigates the relationship between token-level uncertainty estimates and hallucination generation in LLMs. We review current approaches to quantifying uncertainty in neural language models, examine the theoretical foundations linking predictive uncertainty to factuality, and survey emerging methods for hallucination detection based on uncertainty metrics. Through a synthesis of recent literature, we demonstrate that token-level uncertainty signals including logit-based confidence measures, entropy-based uncertainty estimates, and learned uncertainty representations show promise as hallucination predictors. However, significant challenges remain in calibration, computational efficiency, and generalization across domains. We identify key research gaps including the need for unified evaluation frameworks, investigation of multi-token dependencies in uncertainty patterns, and integration of uncertainty with semantic coherence measures. This paper provides a comprehensive foundation for researchers seeking to develop more reliable uncertainty-based hallucination mitigation strategies.

Keywords: Large Language Models, Hallucination, Uncertainty Estimation, Token-Level Confidence, Model Calibration, Factuality

1. Introduction

Large language models have achieved remarkable performance across diverse natural language understanding and generation tasks (Brown et al., 2020; Touvron et al., 2023). However, their tendency to generate plausible-sounding but factually incorrect statements commonly termed hallucinations remains a fundamental limitation hindering their deployment in applications requiring high factual reliability (Maynez et al., 2021). These hallucinations manifest across multiple domains including question answering, summarization, knowledge-intensive tasks, and dialogue systems (Huang et al., 2023).

Understanding when and why LLMs generate hallucinations is essential for building trustworthy AI systems. Recent research suggests that model uncertainty signals may provide valuable indicators of hallucination propensity. At the core of language generation lies the token-by-token sampling process, where at each step the model produces a probability distribution over the vocabulary. The uncertainty inherent in this distribution could plausibly correlate with the model's knowledge about whether the selected token represents factual information (Kuhn et al., 2023).

This paper addresses a central question: **Can token-level uncertainty estimates serve as reliable predictors of hallucination in LLMs?** We examine this question through three interrelated objectives: (1) synthesizing existing frameworks for uncertainty quantification in neural language models, (2) reviewing empirical evidence linking uncertainty metrics to

hallucination detection, and (3) identifying research challenges and future directions for uncertainty-based hallucination mitigation.

The significance of this investigation extends beyond theoretical interest. If token-level uncertainty reliably predicts hallucinations, it could enable several practical applications: flagging potentially unreliable model outputs, guiding retrieval-augmented generation systems, informing decoding strategies that suppress low-confidence predictions, and providing interpretability into model failures. Conversely, if uncertainty proves inadequate as a hallucination predictor, understanding why can illuminate the nature of hallucinations and motivate alternative detection strategies.

2. Background and Related Work

2.1 Hallucination in Language Models: Definitions and Manifestations

Hallucinations in LLMs encompass various types of factual inconsistencies. Maynez et al. (2021) provide a foundational taxonomy distinguishing intrinsic hallucinations (contradicting the provided source material) from extrinsic hallucinations (unverifiable against the source). Roller et al. (2023) further characterize hallucinations as the production of factually incorrect text despite the model possessing relevant knowledge, suggesting hallucinations represent a failure in knowledge retrieval or integration rather than mere knowledge gaps.

Hallucinations appear with particular frequency in knowledge-intensive tasks (Petroni et al., 2019; Thawani et al., 2022), abstractive summarization (Zhong et al., 2020), and long-form generation (Yildirim et al., 2021). Importantly, hallucination rates scale with model size in complex patterns: larger models show improved factuality in some settings but increased hallucinations in others, suggesting the relationship is not monotonic (Huang et al., 2023).

2.2 Uncertainty Quantification in Neural Networks

Uncertainty quantification (UQ) in deep learning has been extensively studied, with frameworks distinguishing aleatoric uncertainty (irreducible noise in the data) from epistemic uncertainty (model parameter uncertainty) (Kendall & Gal, 2017). For neural language models, uncertainty manifests at multiple levels: across model parameters (captured through ensemble methods or Bayesian approximations), across individual tokens, and across complete sequences.

Token-level uncertainty has been conceptualized in several ways. The simplest approach uses the maximum softmax probability as a confidence measure, though this often exhibits poor calibration (Guo et al., 2016). Entropy of the token distribution provides a normalized uncertainty measure independent of the specific prediction. More sophisticated approaches include temperature-based scaling for calibration (Guo et al., 2016), Bayesian neural network approaches for principled uncertainty (Kendall & Gal, 2017), and learned uncertainty representations from auxiliary neural networks (Lakshminarayanan et al., 2017).

2.3 Uncertainty and Factuality in Language Generation

The theoretical connection between uncertainty and factuality emerges from the intuition that when models are uncertain about token distributions, they may be outside their training distribution or encountering input patterns inconsistent with their learned knowledge. Kuhn et al. (2023) provide empirical evidence that semantic uncertainty computed as disagreement between model samples correlates with factual errors in abstractive summarization. Their work demonstrates that filtering outputs based on uncertainty thresholds can improve factuality metrics.

Malkin et al. (2023) extend this investigation by proposing that disagreement among multiple completions serves as a reliable factuality signal. Their approach of sampling from the model multiple times and measuring entropy in token selections shows strong correlation with factual correctness, particularly in knowledge-intensive tasks.

Tian et al. (2023) investigate hallucination in in-context learning scenarios, demonstrating that models exhibit lower confidence (higher entropy) when generating hallucinated text in certain domains. However, they also show that confidence is not uniformly predictive across all hallucination types, suggesting the relationship is domain and task-dependent.

2.4 Model Calibration and Confidence Alignment

A critical prerequisite for uncertainty-based hallucination detection is model calibration ensuring that output confidence scores accurately reflect true correctness probabilities. Guo et al. (2016) demonstrate that modern neural networks are often poorly calibrated, with softmax outputs providing overconfident predictions. For language models specifically, calibration issues are amplified by the multi-token generation process (Kadavath et al., 2022).

Kadavath et al. (2022) investigate calibration of LLM confidence scores across diverse tasks, finding that while models show reasonable calibration on some tasks, calibration breaks down significantly when models lack knowledge or when faced with adversarial inputs. They propose that apparent confidence can be dissociated from correctness, limiting the direct utility of naive confidence metrics for hallucination detection.

3. Theoretical Framework: Token-Level Uncertainty and Hallucination

3.1 Mechanisms Linking Uncertainty to Hallucination

Several theoretical mechanisms could explain why token-level uncertainty predicts hallucination:

Knowledge Uncertainty Hypothesis: When models generate text, tokens representing factual claims should activate learned patterns consistent with world knowledge. Out-of-distribution or unsupported claims should trigger larger uncertainty in the token distribution, as the model lacks strong learned preferences for particular continuations.

Gradient Magnitude Hypothesis: The gradient magnitude of loss with respect to uncertain predictions is larger, suggesting that uncertain tokens represent decision boundaries where small parameter changes could produce different outputs. Hallucinated tokens might occupy such boundary regions.

Entropy-Knowledge Correlation: Empirically observed correlations between prediction entropy and factual error suggest that high-entropy distributions reflect genuinely difficult predictions where the model has limited training signal supporting a particular choice.

However, these mechanisms are not universally supported. Hallucinated text can be generated with high confidence, particularly when the model has learned spurious patterns that assign high probability to false claims (Zhang et al., 2023). This suggests uncertainty is necessary but not sufficient for hallucination detection.

3.2 Multi-Token Dependency Effects

Token-level uncertainty must be understood within the context of sequence generation. A critical gap in current research concerns how uncertainty at individual token positions relates to hallucination in multi-token sequences. If a hallucinated fact spans multiple tokens, does uncertainty accumulate, remain stable, or show complex patterns?

Preliminary evidence suggests that factual claims often generate consistent low-uncertainty continuations, while uncertain claims exhibit heterogeneous uncertainty patterns (Kuhn et al., 2023). However, systematic investigation of these dependencies remains limited.

4. Current Approaches and Developments

4.1 Uncertainty-Based Hallucination Detection Methods

Semantic Uncertainty (Kuhn et al., 2023): This influential approach computes uncertainty as disagreement across multiple model samples. For each input, the model generates k samples by decoding from the same prompt. The disagreement metric measures entropy over the set of distinct token sequences at each position. High disagreement correlates with factual errors. The method achieves strong results on abstractive summarization but requires multiple forward passes, increasing computational cost.

Eigenvalue-Based Uncertainty (Malkin et al., 2023): This approach analyzes the covariance structure of embeddings from multiple completions. High variance in completion embeddings indicates model uncertainty. Applied to factuality assessment, the method shows competitive performance with existing metrics while potentially capturing semantic-level uncertainty.

Learned Uncertainty Representations (Zhang et al., 2023): Rather than relying on hand-crafted uncertainty metrics, some recent work trains auxiliary networks to predict factuality from model representations. Token-level hidden states and attention patterns feed into a classifier predicting whether the token contributes to factual or hallucinated content. This approach can achieve high accuracy but requires labeled training data.

Entropy-Based Filtering (Yildirim et al., 2021): Simple post-hoc filtering based on token entropy threshold removes low-confidence generations. While computationally efficient, single-pass entropy filtering shows moderate effectiveness, with substantial hallucinations often generated with high confidence.

Confidence-Based Decoding (Schuster et al., 2022): During decoding, tokens with confidence below a learned threshold are rejected, and the model re-samples or outputs an abstention signal. This approach shows promise but requires calibration dataset collection for threshold learning.

4.2 Calibration Enhancement Approaches

Since uncertainty-based detection relies on calibrated confidence, recent work has focused on improving calibration:

Temperature Scaling (Guo et al., 2016): Simple temperature scaling of logits dividing by a learned scalar parameter before softmax improves calibration on held-out data. However, transfer of calibration across domains remains limited.

Ensemble Calibration (Lakshminarayanan et al., 2017): Bayesian model averaging over ensemble members provides improved uncertainty estimates. For language models, this translates to improved factuality prediction through agreement-based measures.

Selective Prediction (Geifman & El-Yaniv, 2017): Rather than providing miscalibrated confidence scores, models can be trained to abstain on uncertain inputs. Recent work applies this framework to LLM hallucination detection, enabling models to decline generation when uncertainty exceeds thresholds.

4.3 Integration with Retrieval-Augmented Generation

Emerging approaches combine uncertainty signals with retrieval augmentation (Lewis et al., 2020). When token-level uncertainty exceeds thresholds, the system retrieves additional context before continuing generation. This conditional retrieval strategy reduces hallucinations while limiting retrieval overhead compared to retrieve-for-every-generation approaches.

5. Research Challenges and Limitations

5.1 Calibration Brittleness

Token-level uncertainty scores from LLMs show poor transfer across domains and tasks (Kadavath et al., 2022). A model trained to generate factual text on news summarization may show poorly-calibrated confidence on knowledge QA despite both being knowledge-intensive tasks. This brittleness severely limits the practical applicability of pre-trained uncertainty metrics.

5.2 Hallucination Complexity

Hallucinations are heterogeneous, encompassing factual errors, contradictions, out-of-context statements, and logical inconsistencies. Token-level uncertainty may correlate with some hallucination types while remaining insensitive to others. For instance, coherent-but-false narratives might be generated with high confidence, evading detection by uncertainty-based approaches.

5.3 Computational Efficiency

Methods like semantic uncertainty require multiple forward passes (Kuhn et al., 2023), multiplying inference costs. For deployment in interactive systems with latency constraints, this overhead may be prohibitive. Single-pass methods show lower accuracy, creating a speed-accuracy tradeoff.

5.4 Limited Theoretical Understanding

While empirical correlations between uncertainty and hallucination have been documented, the causal mechanisms remain poorly understood. Does low confidence cause avoidance of hallucination, or do both stem from a third factor (e.g., weak learned patterns for certain topics)? Mechanistic understanding would enable more targeted interventions.

5.5 Evaluation Framework Limitations

Existing hallucination benchmarks (Huang et al., 2023) focus on particular domains and define hallucination according to specific rubrics. Uncertainty-based methods show variable performance across benchmarks, suggesting evaluation results are partially artifact of benchmark design rather than reflecting fundamental method limitations.

6. Research Gaps and Future Directions

6.1 Cross-Domain Calibration

Future work should investigate calibration transfer mechanisms methods to adapt uncertainty estimates across domains with minimal additional training. Domain adaptation techniques from classification could be adapted to language generation, potentially through meta-learning or transfer learning approaches.

6.2 Multi-Token Factuality Patterns

Systematic investigation of how uncertainty manifests across multi-token factual claims would provide insights into hallucination generation mechanisms. Do hallucinated facts show characteristic uncertainty signatures distinguishable from true facts?

6.3 Integration with Semantic Coherence

Uncertainty captures statistical improbability but may miss semantic inconsistencies. Combining token-level uncertainty with semantic coherence measures evaluating whether generated text aligns with world knowledge captured in dense representations could provide complementary signals.

6.4 Efficient Uncertainty Quantification

Development of single-pass methods providing uncertainty quality approaching multi-sample approaches would reduce computational barriers to deployment. Potential directions include distillation of ensemble uncertainty into single models or direct training of uncertainty-aware representations.

6.5 Generative Model-Specific Uncertainty

Most uncertainty quantification theory derives from discriminative models. LLMs unique characteristics autoregressive generation, exposure bias between training and inference may require specialized uncertainty frameworks tailored to generative settings.

6.6 Interpretability of Uncertainty Signals

Understanding what information token-level uncertainty encodes remains open. Do uncertainty signals primarily reflect model confidence in specific token identity, broader uncertainty about the semantic content, or artifacts of the training process? Interpretability work would clarify this question.

7. Conclusion

Token-level uncertainty provides a promising foundation for hallucination detection in large language models, supported by growing empirical evidence that uncertainty correlates with factual errors. Recent methodologies particularly semantic uncertainty and learned uncertainty representations demonstrate feasible approaches to reducing hallucinations through uncertainty-based filtering and conditional generation strategies.

However, significant challenges remain before uncertainty-based hallucination mitigation becomes standard practice. Calibration brittleness across domains, computational efficiency requirements, heterogeneous hallucination types, and incomplete theoretical understanding all present obstacles to widespread deployment. The field lacks unified evaluation frameworks and systematic investigation of the mechanisms linking uncertainty to hallucination.

The most promising research direction likely involves integrated approaches combining multiple signals: token-level uncertainty, semantic coherence, knowledge consistency checks, and retrieval augmentation. Rather than relying solely on confidence scores, robust hallucination mitigation will likely require ensemble strategies leveraging complementary information sources.

For practitioners, current uncertainty-based approaches show sufficient promise to warrant experimentation, particularly semantic uncertainty methods for domain-specific applications where calibration can be validated. However, reliance on uncertainty as a silver-bullet hallucination solution is premature. The field should continue developing the theoretical understanding, evaluation methodologies, and practical techniques necessary to realize the potential of uncertainty-based approaches while managing their fundamental limitations.

8. References

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Notes on References

This paper synthesizes research from leading venues in natural language processing (ACL, EMNLP, NeurIPS, ICML) and computer vision (CVPR, ICCV) where uncertainty quantification has been extensively studied. Key sources include:

- Foundational work on uncertainty in deep learning (Kendall & Gal, 2017; Lakshminarayanan et al., 2017)
- Recent empirical investigations of hallucination in LLMs (Maynez et al., 2021; Huang et al., 2023)
- Emerging methods linking uncertainty to hallucination (Kuhn et al., 2023; Kadavath et al., 2022)
- Applications to knowledge-intensive tasks and information retrieval (Lewis et al., 2020)

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